

1. Solution Analysis

This document presents the solution analysis and system design phases for the proposed SME Operations Management System. The objective of the solution is to improve operational efficiency within a small business environment offering food, printing, and pool table services through the implementation of a structured digital management system.

1.1 Introduction

This section evaluates possible digital solutions to address the identified challenges. AI tools were used to generate and refine these solutions, which were then analysed to determine the most suitable option.

1.2 AI Tools Used

Possible solutions were explored using AI-assisted ideation tools including ChatGPT, Microsoft Copilot, and Google Gemini. Each tool provided different perspectives, which were critically evaluated and refined to suit the SME's needs.

1.3 Identified Solutions

The first solution is an integrated web-based management system that centralises all business operations into a single platform. This system allows staff to manage food orders, printing jobs, pool table usage, and financial transactions in one place. It reduces errors, improves tracking, and enhances overall efficiency. However, it requires internet connectivity and initial setup. The second solution is a mobile-based POS system that allows staff to manage operations using smartphones or tablets. This solution improves real-time tracking and flexibility but may require compatible devices and consistent usage by staff.

The third solution involves a combination of low-cost tools such as a basic POS system, timer applications, and Google tools. This option is affordable and easy to implement but lacks full integration and requires manual coordination.

1.4 Comparison of Solutions

The three solutions were compared based on cost, ease of implementation, level of integration, and suitability for the SME.

The **integrated web-based system** provides the highest level of integration, as all business functions are managed within a single platform. However, it may require more effort to develop and maintain. The **mobile-based POS system** offers flexibility and ease of use, allowing staff to manage operations using portable devices, but may involve additional costs for devices and software features. The **low-cost combination solution** is the most affordable and easiest to implement, but it lacks full integration and may result in inefficiencies due to the use of multiple separate tools.

1.5 Recommended Solution

The integrated web-based management system is recommended as the most suitable solution for the SME. This is because it provides a centralised platform that addresses all key operational challenges in a structured and efficient manner. It improves accuracy, reduces manual errors, and enhances overall control of business activities.

Although the initial setup may require some effort, the long-term benefits in efficiency, organisation, and scalability outweigh the limitations. This solution also supports future growth, as additional features can be integrated as the business expands.

1.6 Value and Context

Solution	Benefits	Context
Integrated Web-Based System	Improves efficiency, reduces errors, centralises all operations, supports better decision-making	Suitable for SMEs seeking a long-term, scalable solution
Mobile-Based POS System	Easy to use, flexible, improves transaction tracking, supports mobility	Suitable for SMEs with access to mobile devices and moderate budget

Low-Cost Tools Combination	Affordable, easy to implement, improves basic tracking and record-keeping	Suitable for small businesses with very limited budget and minimal technical resources
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2. Design

2.1 Introduction

This section presents the design of the proposed digital solution for the SME. The design is based on the problems identified during requirements gathering and analysis, particularly the lack of structured systems for managing food orders, printing services, and pool table usage. The aim is to design a simple, affordable, and effective system that improves daily operations while remaining suitable for a small business environment.

2.2 System Overview

The proposed system is a simple digital management system that integrates the core services of the business into one platform. The system will allow staff to manage food orders, record printing jobs, manage pool table usage, and track daily income in real time.

The system will replace manual processes such as paper-based order taking and informal tracking methods. It will provide a central point where all business activities are recorded and monitored, reducing errors, improving service delivery, and supporting better decision-making.

2.3 Technology Architecture

The system will follow a simple client-server structure suitable for a small business. Staff will use devices such as smartphones, tablets, or a computer to access the system through a web-based interface or lightweight application. The system will require an internet connection for data storage and synchronization, especially if cloud storage is used. A central database will store all information related to orders, printing jobs, pool usage, and transactions.

Basic network infrastructure such as a Wi-Fi router will connect all devices within the business. The system will be designed to run on affordable hardware to accommodate the financial limitations of the SME. The architecture also supports future database integration for structured transaction storage and reporting capabilities.

2.4 Functional Modules

The system will be divided into key functional modules based on the business services:

Order Management Module:

This module will allow staff to capture and manage food orders digitally. Orders will be recorded with details such as item type and quantity, reducing the chances of lost or incorrect orders.

Printing Management Module:

This module will record all printing requests, including customer details, file source, and job status. It will help track pending and completed jobs, ensuring better organisation and accountability.

Pool Table Management Module:

This module will manage pool table usage by tracking game sessions or time. It will help reduce disputes between customers and ensure fair usage of the service.

Payment and Reporting Module:

This module will record all transactions from the different services and generate daily reports. It will improve financial tracking and provide insights into business performance.

2.5 User Interface Design

The system interface will be designed to be simple and easy to use for staff with basic technical skills. Each service will have a clear screen or section, such as an order screen for food, a print job screen, and a pool tracking screen.

The interface will focus on clarity, with minimal steps required to complete tasks, ensuring efficiency during busy periods.

2.6 Security, Privacy and Ethical Considerations

The system will restrict access to authorised users and ensure that customer data is handled responsibly. It will promote fairness and transparency in transactions and service delivery.

2.7 Conclusion

The proposed solution provides a structured and scalable approach to improving operational efficiency within the SME environment. By integrating multiple business services into a centralised system, the solution addresses key operational challenges while supporting future growth, improved service delivery, and better business monitoring.